

1.3) Richard Stallman's philosophy and the GNU project:

➤ Who is Richard Stallman?



Let's delve into the story of a key figure in the world of computing: Richard Stallman, one of the pioneers of the free software movement.

If the term "free software" is widely known today, it's largely thanks to his commitment and ideas.

Richard Matthew Stallman, often called (RMS), was born on March 16, 1953, in New York City. From a very young age, he developed a great passion for mathematics and computer science.

Endowed with exceptional curiosity and a remarkable analytical mind, he quickly distinguished himself through his talent. But it was truly during his studies at Harvard University that his path in the world of computing took a decisive turn.

It was from this period that the adventure began that would make him one of the most influential figures in the history of free software.

After graduating from Harvard University, Richard Stallman joined the Artificial Intelligence Laboratory at the Massachusetts Institute of Technology, better known as MIT.



At that time, the culture of programmers was based on sharing and collaboration. Researchers freely exchanged their programs and ideas to improve computer systems. For Stallman, this way of working represented the very spirit of computing.

However, in the early 1980s, the situation gradually changed. Companies began to close the source code of their software and prohibit its modification or redistribution. For Stallman, this development was unacceptable because it deprived users of their freedom.

➤ **Birth of the GNU project (1983):**

Richard Stallman and the Printer Incident:

In the early 1980s, at the Artificial Intelligence Laboratory of the Massachusetts Institute of Technology (MIT), Stallman was using a network printer.

The lab's old printer had a driver whose source code was accessible: Stallman had been able to modify it to send a notification to users when their print jobs were blocked (paper jam, etc.).

With the new printer, the driver's source code was no longer accessible (proprietary software).

Therefore, Stallman could not modify the program to correct the problem of frequent blockages.

He tried to obtain the source code from another university, but was denied access due to a confidentiality agreement.

This event deeply affected Stallman.

He saw this situation as a loss of freedom for both users and programmers. In response to this situation, he launched the GNU project in 1983, whose goal was to create a completely free operating system.



Richard Stallman et le gnou, emblème du projet GNU

The goal of "GNU's Not Unix" is to create a completely free Unix-like operating system. Unix had existed since 1969, but it was proprietary. A few years later, a significant event would transform the history of free software.

In 1991, a Finnish student named Linus Torvalds announced on the internet that he was working on a new operating system kernel. He called this project Linux.

Initially, this project was modest. But very quickly, programmers from all over the world began to contribute. Thanks to the internet, they collaborated, fixed bugs, and improved the system.

By combining the Linux kernel with the tools developed in the GNU project, developers created a complete operating system called GNU/Linux.

For the first time, it became possible to use a completely free operating system that anyone could study, modify, and share.

This model of open collaboration marked a true revolution in the world of computing.

Today, the GNU/Linux system is used everywhere:

On internet servers, in data centers, in supercomputers, and even in smartphones.

A few years later, in 1985, he founded the Free Software Foundation (FSF), an organization dedicated to promoting and protecting free software.



The FSF's mission is to support the development of free software and promote the idea that software should respect users' freedom.

Stallman's advocacy doesn't stop at code.

He strongly opposes software patents, DRM (Digital Rights Management), and any form of digital restriction.



Richard Stallman states: “Proprietary software is a social injustice.” Software is free if it respects these freedoms:

- 0- Freedom to use the program for any purpose
- 1- Freedom to study how it works (access to the source code)
- 2- Freedom to modify the program
- 3- Freedom to redistribute copies (modified or unmodified)

Without these freedoms, the software is not free, even if it is free of charge.

To legally guarantee these freedoms, Stallman also created the GNU General Public License, better known as the GPL. This license ensures that free software remains free, even after modification or redistribution.

Beyond his technical contributions, Stallman's work sparked crucial debates about ethics in the technology industry.

He showed us that technology is not just about innovation, but also about principles and its impact on society.

The history of free software demonstrates that another way of developing technology is possible.

Thanks to pioneers like Richard Stallman and Linus Torvalds, knowledge sharing and global collaboration have become essential pillars of modern computing.

Today, free software plays a strategic role in technological innovation and continues to influence the future of digital technology.

Richard Stallman, founder of the free software movement, visited Algeria from February 2nd to 13th, 2015, giving a series of lectures on the philosophy of free software and digital ethics. His presentations took place at several universities (Sétif, Sidi Bel Abbès, Djelfa, and Saïda) to promote the use of free software and combat digital surveillance.

